

Summary: 2-Data Fundamentals for BI - Part1

Generated on June 25, 2026

- Data Fundamentals for Business Intelligence (BI)
- Understanding data in the context of business operations
- Data types, sources, and quality: • Data governance and compliance considerations for business • Data Quality: Why Preprocess the Data? - Measures for data quality (checking if information is good or bad): a multidimensional view
- - Accuracy: correct or wrong, accurate or not accurate, completeness: not recorded, unavailable, ... Consistency: some modified but some not, dangling,.. Timeliness: timely update? Believability: how trustable the data are correct? Interpretability: How easily the data can be understood? Data Cleaning: Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- - Data integration: Integration of multiple databases, data cubes, or files
- Data reduction: Dimensionality reduction, Numerosity reduction, data compression, data transformation and data discretization
- - Normalization: Concept hierarchy generation
- - Concept hierarchy generation: concept hierarchy generation, conceptualization, and conceptualization
- - Basic Concept: Basic Concept of Business Intelligence:
- - Main Concept: Business Intelligence in Business Operations
- - Key Concepts: Business Operations, Operations, Operations
- - Business Processes
- - Operations and Operations: Operations, operations, operations
- - Operations and operations: Operations and operations
- • Data Governance and Compliance considerations:
- Big Data: Data governance: Data quality: why preprocess the data and how it is processed
- - How it's processed and processed
- Data Cleanup: filling in missing data, cleaning data, integration, data integration, and data compression
- Data Integration: Data compression: Data transformation: data compression and data transformation
- - data transformation, data normalization, data distribution
- • Topic: Data Cleaning
- Basic Concept: Noisy Data: random error or variance in a measured variable makes the data less clear and harder to analyze
- Incorrect attribute values may be due to faulty data collection instruments or faulty data entry problems
- inaccurate data transmission problems: technology limitation, inconsistency in naming convention, lack of naming convention. How to handle noisy data: • Binning: first sort data and partition into (equal-frequency) bins
- • Smoothing by bin means: Replace each age in the - bin with the average age of that bin (which would be around)
- Smoothed by bin median: Replace the middle age of the -bin with the middle of the other bins (either or). • Clustering: Groups similar data points together: detect and remove outliers (are often far away from the clusters)
- • Regression: fit the data into regression functions: detect, remove, and deal with outliers
- • Clustered Data: groups similar data point together; detects and removes outliers; deals with possible outliers. • Data cleaning as a Process: like detective work to find errors and then fixing them: detect suspicious values and

Summary: 2-Data Fundamentals for BI - Part1

Generated on June 25, 2026

check by human (e.g., deal with possible outbursts)

- Data clean-up as a process: detecting suspicious values, checking by human, dealing with possible Outbursts: detecting and dealing with Possible Outbursts: dealing with outbursts. • Topic: Data Cleaning and Data Integration
- Basic Concept: Data cleaning is like detective work to find errors and then fixing them. Data integration combines data from multiple sources into a single store. Data migration and integration: moving and combining data from different sources. Data reduction: Dimensionality reduction, Numerosity reduction, Data compression. Data transformation and data discretization: Normalization, Concept hierarchy generation, Data integration, Schema integration, and Data Scrubbing. • Data Preprocessing:
- Fill in missing values, smooth noisy data, identify or remove outliers, resolve inconsistencies, and resolve inconsistencies (e.g., postal code, spell-check)
- Data Processing:
- Normalization: fill in missing values, smooth data, identifying or removing outliers, resolving inconsistencies, normalizing data, reducing dimensionality, reducing noise, reducing error rates. Data Analysis: analyze data to discover rules, relationship to violators, correlation and clustering to find outliers. • Data Integration: Combines data from several sources into one store. - Data integration: matching up the structure of the data from various sources. - Data migration tools: transfer data and make changes during transformation. - ETL (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface. • Topic: Data Integration and Redundancy
- Redundant data occur often when integration of multiple databases, data cubes, or files. Careful integration of the data from multiple sources improves speed and quality. • Redundants can be detected by correlation analysis (how strongly two things are related) and Covariance Analysis (how two things change together)
- Data integration: Combines data from several sources into a coherent store. • Correlation Analysis (Nominal Data): A statistical test that helps us understand if there's a relationship between two categorical variables (variables that can be divided into categories, like colors, types of fruit, or survey responses)
 - - The larger the χ^2 value, the more likely the variables are related (Strong)
- Redundancies in Data Integration:
- Basic Concept:
- Concept hierarchy generation: concept hierarchy generation
- Concept structure generation: Concept hierarchy generation: concept structure generation
- Data integration (Schema Integration): Schema integration: matching up the structure of data from different sources. • Data reduction: Dimensionality reduction: Numerosity reduction and data compression: Data transformation and data discretization: Data compression, data transformation, normalization, and data normalization
- Data transformation: Normalization: normalization of data into a consistent, coherent store
- Documenting: Documenting and resolving data value conflicts: For the same real-world entity, attribute values from different source are different. E.g., one database might say a customer's age is , while another database says it's . • Derivable data: One attribute may be calculated (derived) from another attribute in another database, e.g. annual revenue data: one database says annual revenue, another says age. • Correlation analysis: correlation analysis (chi-square) test: is a statistical test for relationship between variables (categorical

variables). - Correlation does not imply causality: correlation does not mean one causes the other (correlated, it doesn't mean one cause the other.)

- #
- • • Basic Concept: Correlation is a statistical measure that expresses the extent to which two variables are linearly related, meaning how much they change together at a constant rate. • Correlation does not imply causality (Just because two things are related (correlated), it doesn't mean one causes the other.) • Positive Correlation ($r > 0$): As one variable increases, the other tends to increase. • Example: Height and weight are correlated ($r = 1$)
- Correlation Analysis (Numeric Data): • Positive Correlation ($r < 1$): • One variable increases as the other decreases. • Positive correlation ($n > 1$): • Two variables increase as the number of tuples increases. • Negative correlation ($p > 1$: $n = 1/2$)
- • Negative correlations ($p = 0.1/2$): • Both variables are causally linked to the third variable: population
- Chi-Square Calculation: An Example:
- Numbers in parenthesis are expected counts calculated based on the data distribution in the two categories. It shows that like_science_fiction and play_chess are correlated in the group. • Redundancy Analysis: Redundancies (Redundancy) - Redundant variables that are not related to each other at all. • Redundance Analysis (Unexplained): Redundants that are related to more than one variable. • • Basic Concept: Correlation measures the linear relationship between two variables. • Correlation Analysis (Numeric Data)
- Positive Correlation ($r > 0$): As one variable increases, the other tends to increase. • Negative Correlation ($r < 0$): as one variable decreases, other decreases. • Visually Evaluating Correlation:
- Scatter plots showing the similarity from - to . The closer r is to - or , the stronger the linear relationship. • Handling Redundancy: Covariance (N numeric data)
- Covariant measures how two variables change together. • Correlation coefficient (BAB):
- BAB measures the correlation between A and B. • Average Correlation Coefficient (AAC):
- AAC is the average correlation coefficient of AAC and BAC. • Normal Correlation coefficient (AEC):
- Average correlation coefficient is the standard deviation of A AC and B AC. • Natural Correlation Coefficient:
- AEC is the correlation coefficient between BAC and AAC. • Visually evaluating Correlation • Redundancy: Covariance (Numeric Data)
- If $CovA, B > 0$ then A is greater than its average, B also tends to be greater than its average, and vice versa. • Positive CovA: Height and weight might have positive covariance; temperature and hot chocolate sales might have negative covariance. • Independence: Knowing the value of A doesn't tell you anything about B; shoe size and IQ score likely have zero (or very close to zero) • Co-Variance: An Example: Suppose two stocks A and B have the following prices in one week:
- Co-variance: There's no linear relationship. Knowing the value of A does not tell you anything about B. • Data Cleaning: Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies. Data integration: Integration of multiple databases, data cubes, or files. Data reduction: Dimensionality reduction, Numerosity reduction, Data compression, Data transformation and data discretization. Data normalization: Normalization, Normalization and Data Discretization - Concept hierarchy generation. • Topic: Data Preprocessing and Data Reduction
- Basic Concept: Data preprocessing involves filling in missing values, smooth noisy data, identify or remove

outliers, and resolve inconsistencies. • Data Reduction: Obtain a reduced representation of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results

- Data Reduction Techniques:
 - - Numerosity reduction (some simply call it: Data Reduction) - Regression and Log-Linear Models - Histograms, clustering, sampling - Data cube aggregation - Data compression
 - - Wavelet transforms (Principal Components Analysis)
 - - Principal Components Analysis (PCA) - Supervised and nonlinear techniques (e.g., feature selection, feature creation, noise frequency reduction)
 - - Two Sine Waves + Noise Frequency - Noise Frequency Transform (2SW + 2F) - Noise frequency transform (2F + noise frequency) - Normalization
 - - Concept hierarchy generation
- Data Reduction: Dimensionality Reduction
 - - Avoid the curse of dimensionality
 - - Help eliminate irrelevant features and reduce noise
 - - Reduce time and space required in data mining
 - - Allow easier visualization
- Data reduction strategies: Wavelet transforming, principal components analysis, data transformation, data discretization. • Topic: Wavelet transform (DWT)
- Basic Concept: Wavelets are mathematical techniques that decompose data (signal) into different frequency components, to identify and discard less important components. They are powerful for analysing signals that have sudden changes or discontinuities. The DWT can focus on the most important parts of the signal and discard the rest, resulting in smaller file sizes without significantly sacrificing quality. • How the DWT Works:
- Padding: The length of the data needs to be a power of 2. If it's not, add zeros to the end (padding). Smoothing & Difference Functions: The DWT uses two special functions: one that "smooths" the data (selection) and another that "makes the difference"
- - Haar2 Daubechies4: Haar is a technique used to map data to a new space
- Two Sine Waves + Noise Frequency
 - - Noise Frequency: noise frequency divided by two sine waves
 - - Sine waves are localized waves of different scales. - Noise frequency is the frequency at which a particular noise occurs
 - - Two sine wavelets are localised waves with different frequencies
 - - Wavelet Transform represents the signal in terms of these wavelets, specifying both the frequencies present and when (or where) they occur. • Example:
- Image: Wavelets are excellent for image compression (like JPEG). They can identify sharp changes in an image (like edges) very efficiently. • Audio: If you have a recording with a sudden loud noise, wavelets pinpoint exactly when that noise occurred. • Medical: wavelets used in analyzing EEG data, where sudden spikes or changes in the signal can be important. • Topic: Discrete Wavelet Transform (DWT)
- Basic Concept: DWT is a technique used to decompose a signal (like a sound or image) into different frequency components. The Haar wavelet is a specific type of wavelet that is used in the DWT. The DWT uses two special functions: one that "smooths" the data (averaging nearby values) and one that calculates the "difference"

Summary: 2-Data Fundamentals for BI - Part1

Generated on June 25, 2026

between nearby values (like highlighting changes)

- Details: This indicates the level of detail.
- Averages: These represent the smoothed or averaged values of the data at each resolution.
- Detail Coefficients: These capture the details or fluctuations that were lost when smoothing the data.
- Difference: The "differences" or "details" lost in the smoothing process:
- Detail Coefficient: $(+/-) = (+/-) = + / - / + / - / + / - / + / - / - / - /$.
- Wavelet Decomposition:
- Starting Data (Resolution): data points (as measurements or samples of a signal). e.g., sound wave, temperature readings).
- First Transformation (resolution): $[, , ,]$
- Second Transformation (Resolutions): $[, ,]$.
- Third Transformation (Resolution) : $[,]$.
- Fourth Transformation (resolution): $[,]$.
- Fifth Transformation (resolution: $[,]$)
- Sixth Transformation (Resolution: $[,]$)
- Seventh Transformation (resolutions: $[,]$)
- Eighth Transformation (resolving: $[,]$)
- Ninth Transformation (olution: $[,]$)
- Topic: Hierarchical Decomposition (Haar transform) - decomposing a signal into different frequency components
- Basic Concept: The Haar transform decomposes signals into frequency components. The structure of the Haar transforms is hierarchically based on wavelet decomposition.
- Haar Transform (Resolution): 1. Averages: smoothing function applies a "smoothing" function to pairs of data points and calculates the averages:
- $(+/-) = (+/-) = + / - / = (+/-) / - / - / - / - /$
- Detail Coefficients: $[, , -] - / / / - / , / , , , - , / , - / , / , . - / ,$
- Data Mining: Concepts and Techniques: concepts and techniques:
- - Data Mining Techniques: Concepts, Techniques, Concepts, and Techniques, by Jiawei Han, Micheline Kamber, Jian Pei, Foster Provost, and Tom Fawcett
- - Techniques: Data Mining, Data Science for Business, Data Mining techniques, Data Processing Techniques, Data Analysis Techniques, and Data Mining Methods, by Foster Provst, Tom Pawcett, and Michelin Kamber.